# Olivier Bisson

*PhD Student in Geometric Statistics*

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## Profile

I study the **geometry of smooth manifolds and stratified spaces**, with a focus on correlation and covariance matrices. My work combines **Riemannian and quotient geometry**, **orbit-type stratification**, and geometric statistics. I investigate intrinsic distances, rank-constrained geodesics, curvature, holonomy, and volume measures on spaces of low-rank correlation matrices, alongside log-Euclidean constructions for full-rank matrix manifolds. Applications include statistical modeling of brain connectivity trajectories.

## Research interests

Geometry of stratified spaces and orbit-type quotients; Riemannian submersions and principal bundles; intrinsic metrics and rank-constrained geodesics; curvature, connections, and holonomy; volume measures on matrix spaces; log-Euclidean geometry; geometric statistics and applications to neuroimaging.

## Current position

Oct. 2023 – present **PhD Student in Geometric Statistics**, *Inria (EPIONE) & Université Côte d'Azur*, France
**Thesis:** *Riemannian geometry and orbit type stratification of correlation matrices.*
**Advisor:** Xavier Pennec.
**PhD defense:** scheduled for 12 November 2026.
**Funding:** 3IA (Interdisciplinary Institute for Artificial Intelligence).

## Education

2023 **M.Sc. in Mathematical Engineering**, *Stochastic and Statistical Modeling*, Université Côte d'Azur, France, Highest honors

2022 **M.Sc. in Pure and Applied Mathematics**, *Algebra and Geometry*, Université Côte d'Azur, France

2020 **B.Sc. in Fundamental Mathematics**, Université Côte d'Azur, France

## Published journal papers

2024 **O. Bisson**, X. Pennec. *Differential of the Stretch Tensor for Any Dimension with Applications to Quotient Geodesics*. *Comptes Rendus. Mathématique*, 362 (2024), pp. 1847–1856. doi:10.5802/crmath.692. PDF.

## Conference papers

2025 **O. Bisson**, Y. Aeschlimann, S. Deslauriers-Gauthier, X. Pennec. *Log-Euclidean Frameworks for Smooth Brain Connectivity Trajectories*. In: *Geometric Science of Information (GSI)*, St Malo, France, 2025, pp. 194–204. Springer. doi:10.1007/978-3-032-03924-8_20. arXiv:2507.15374. *Nominated for Best Paper Award.*

## Preprints

2026 **O. Bisson**, D. Loiseaux, X. Pennec. *On the Geometry of Low Rank Correlation Matrices*. arXiv:2610.28475.

2025 **O. Bisson**, X. Pennec. *Log-Euclidean Lie Groups*. arXiv:2511.13380.

## Talks & conferences

Oct. 2026 *On the Geometry of Low Rank Correlation Matrices*. Statistical Foundations for the Analysis of Non-Euclidean Data, Institute for Mathematical and Statistical Innovation (IMSI), University of Chicago, Chicago, USA, 21 October 2026.

Jul. 2026 *Poster: Log-Euclidean Lie Groups and rs-fMRI Brain Connectivity Trajectories*. Foundations of Computational Mathematics (FoCM), Faculty of Mathematics, University of Vienna, Vienna, Austria, 8–18 July 2026.

May 2026 *Log-Euclidean Geometry of Full-Rank Correlation Matrices*. Geometry, Topology and Machine Learning 2026 (GML 2026), Sorbonne Université, Paris, France, 18–22 May 2026.

Dec. 2025 *Mini-course: Quotient Spaces and Related Differential Geometry*. EPIONE Seminar, Inria Sophia Antipolis, France, 12 December 2025.

Oct. 2025 *Log-Euclidean Frameworks for Smooth Brain Connectivity Trajectories*. Geometric Science of Information (GSI), St Malo, France, 29 October 2025. *Nominated for Best Paper Award.*

Mar. 2025 *Geometry, Stratification and Applications of Structured Correlation Matrices*. EPIONE Seminar, Inria Sophia Antipolis, France, 5 March 2025.

Feb. 2025 *Differentiable Structures on Correlation Matrices and Applications in Neuroimaging*. *Infinite-dimensional Geometry: Theory and Applications*, ESI (Vienna), Austria, 10 February 2025.

Dec. 2024 *Geometry, Stratification and Applications of Structured Correlation Matrices*. 3IA Côte d'Azur PhD Seminar, Nice, France, 6 December 2024.

Nov. 2024 *Riemannian Metrics on Correlation Matrices with Applications in Neuroimaging*. Mathematical Imaging and Random Geometry (RT MAIAGES), Nice, France, 25 November 2024.

Oct. 2024 *Introduction to the Riemannian Geometry of Correlation Matrices*. EPIONE Seminar, Inria Sophia Antipolis, France, 18 October 2024.

Feb. 2024 *A framework to put metrics on quotient spaces, with application in medical image analysis*. EPIONE Annual Retreat, Auron, France.

Dec. 2023 *Poster: Geometry, stratification and applications of structured correlation matrices*. 3IA Scientific Day, SKEMA, Sophia Antipolis, France.

Oct. 2023 *A framework to put metrics on quotient spaces, with application in medical image analysis*. EPIONE Seminar, Inria Sophia Antipolis, France.

## Awards & Funding

2023–2026 **3IA Côte d'Azur PhD Funding**, Université Côte d'Azur / Inria.

2025 **Best Paper Award nomination** (GSI 2025) for *Log-euclidean frameworks for smooth brain connectivity trajectories*.

## Teaching

2025–2026 **Teaching assistant (TD)**, *Introduction à l'analyse (L1)*, Université Côte d'Azur
Sets and functions; limits and continuity; differentiability; function study; injective/surjective/bijective maps; integrals and primitives.

2024–2025 **Teaching assistant (TD)**, *Compléments d'algèbre linéaire (L2)*, Université Côte d'Azur
Orthogonality and projections; Gram–Schmidt; least squares; isometries; introduction to PCA.

2024–2025 **Teaching assistant (TD)**, *Calculus II (L2)*, Université Côte d'Azur
Complex numbers; linear ODEs; multivariable calculus (partials, gradient, tangent planes, level sets, local extrema).

2023–2024 **Teaching assistant (TD)**, *Maths Approfondissement 2 (L1)*, Université Côte d'Azur
Exponential/logarithm refreshers; linear ODEs; Riemann integration basics; partial fractions; Euclidean vector geometry.

## Software & open-source

Contributor to the `geomstats` Python package, with implementations related to Riemannian geometry on SPD and correlation matrix manifolds.

## Previous Research experiences

Apr. 2023 – Oct. 2023 **Research intern**, *Inria (EPIONE), Université Côte d'Azur*, Nice/Valbonne, France
**Topic:** *Interpolation and extrapolation of correlation matrices for dynamic functional connectivity*.
**Supervisors:** Xavier Pennec, Samuel Deslauriers-Gauthier.
Implementation and evaluation of Riemannian metrics on correlation matrices for geodesic interpolation and extrapolation on resting-state fMRI correlation time-series.

Sep. 2022 – Feb. 2023 **Research project**, *LJAD & CEPAM, Université Côte d'Azur*, Nice, France
**Topic:** *Supervised classification of wood charcoal fragments: from machine to active learning*.
**Supervisor:** Marco Corneli.
Development of learning methods to automatically identify wood charcoal remains from a Southern-African context (M.Sc. capstone project).

Mar. 2022 – Jun. 2022 **Research intern**, *LJAD, Université Côte d'Azur*, Nice, France
**Topic:** *Quantum representations of mapping class groups*.
**Supervisor:** Jeremy Toulisse.
Topological/combinatorial approach (skein theory) to construct quantum representations of mapping class groups.

## Languages

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|---|---|---|
| French | Native | *Fluent* |
| English | Fluent | *Reading, writing, speaking* |
| Italian | Basic | *Working knowledge* |

## Additional

Outside research: Volleyball, rock climbing, sailing, beach volleyball and swimming. I am deeply convinced that **anyone** can do mathematics.